Enhancing MD-Judge via a Teacher-Student Safety-Judging Framework

1. Motivation

Current MD-Judge models often misclassify borderline harmful prompts, especially in adversarial or multi-turn scenarios. They lack nuanced reasoning and fail to generalize to unseen attack strategies. We propose a Teacher-Student safety-judging framework where a Teacher-Judge LLM (larger, highly aligned, e.g., GPT-4-class) provides feedback, rationales, and distillation signals to train a smaller, efficient Student-Judge (MD-Judge). This approach combines the accuracy and interpretability of the Teacher with the efficiency and deployability of the Student.

2. Method Overview

• Teacher-Generated Labels & Rationales: The Teacher-Judge evaluates QA pairs (from SALAD-Bench, Beavertails, adversarial sets) and outputs not only safe/unsafe labels but also step-wise rationales, e.g., highlighting harmful intent or refusal quality.

• Distillation Phase: Student-Judge (MD-Judge) is fine-tuned on these Teacher-generated rationales + labels using Knowledge Distillation or LoRA-based SFT. The loss combines classification loss and rationale-agreement loss, teaching MD-Judge why a case is harmful.

• Interactive Feedback Loop: In ambiguous or low-confidence cases, Student-Judge queries Teacher-Judge for corrective feedback. We use iterative self-play: Student explains its reasoning → Teacher critiques and corrects → new synthetic data improves the Student.

• Curriculum Learning: Start with easy cases (clear harmful vs safe) → progress to harder adversarial and multi-turn attacks, helping MD-Judge gradually master complexity.

3. Experimental Plan

Datasets: SALAD-Bench (base), Beavertails (harmful refusal), Emoji Attack perturbations, multi-turn jailbreak simulations.
Baselines: Original MD-Judge vs. Teacher-Student MD-Judge.
Metrics: F1-score, false-positive/negative rate, robustness to adversarial perturbations, rationale quality (human-rated).
Ablations: With vs. without rationale distillation; effect of curriculum stages.

4. Expected Benefits

• Higher Accuracy: Better boundary detection for subtle harmful intent.

• Rationale-Aligned Safety: Student learns human-interpretable decision patterns.

• Improved Robustness: Handles adversarial prompts and long-context attacks more reliably.

• Efficiency: Student retains near-Teacher accuracy with much lower inference cost.

6. References

[1] Rui Ye et al., "Emerging Safety Attack and Defense in Federated Instruction Tuning of LLMs," ICLR 2025.

[2] SALAD-Bench: Safety Alignment Dataset and Benchmark — GitHub: https://github.com/OpenSafetyLab/SALAD-BENCH

[3] Emoji Attack: Vulnerabilities in LLM Safety Judges to Unicode Perturbations, arXiv:2411.01077 (2024).

[4] Hinton et al., Knowledge Distillation, NeurIPS 2015.

[5] Ouyang et al., Training Language Models to Follow Instructions with Human Feedback, NeurIPS 2022.